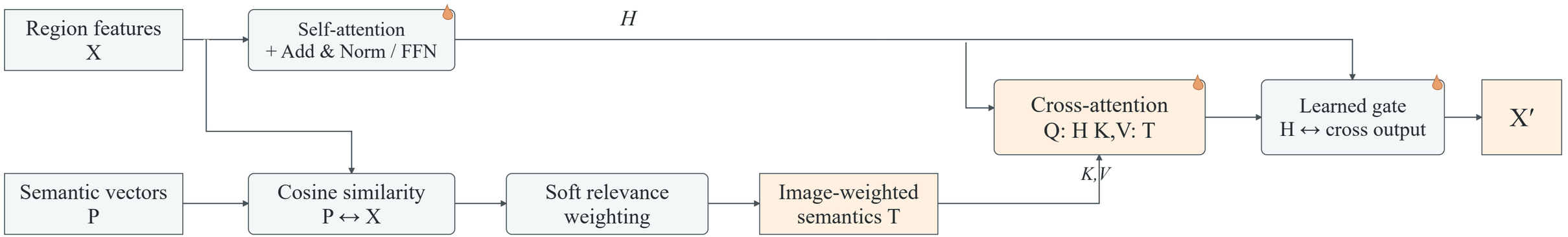
Where visual–semantic alignment happens

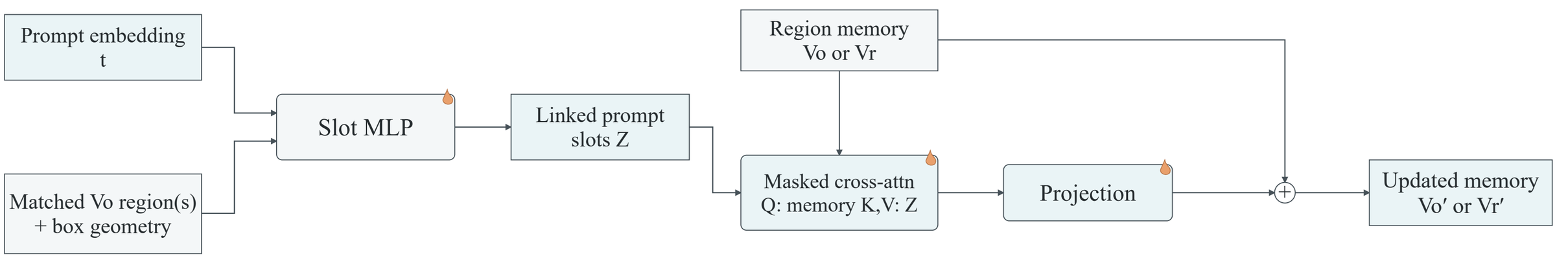
A. Semantic bank–region alignment



Object: (V, Pn) → (V_1, T_1) → (Vo, T_2); relation: (Vo, Pv) → Vr.

Second object stage uses T_1. Standard residual / normalization / FFN steps are abbreviated.

B. Image-specific prompt–region alignment



Attention is allowed only between each ROI and its linked prompt slots; unmatched ROIs stay unchanged.

A learns soft semantic correspondence; B restricts prompt attention using matched object / relation endpoints.